

PAW3311DB-T2MU: Low Power Optical Gaming Navigation Chip

General Description

The PAW3311DB-T2MU is PixArt Imaging's new low-power gaming navigation chip suitable for wired and wireless gaming applications. It has the latest state-of-the-art low-power architecture and automatic power management modes, making it ideal for battery-operated, power-sensitive cordless gaming devices. It provides an excellent gaming experience with high speed and high-resolution features, even in low power mode, to fulfill gamers' needs. It is packaged in an 8-pin staggered dual-in-line package (DIP) and designed to match with LM31-LNG, LM33-LSG, LOAL-LSG1 (5mm IR illumination source), or LOAJ-LSG1 lens (SMT IR Illumination source) to achieve optimum performance.

Key Features

- Low power consumption of typical 2.3mA @ run mode
- Programmable rest modes
- Small form factor PDIP 8L molded lead-frame package
- Operating Voltage: 1.80 to 2.10V
- High speed motion detection 300ips and acceleration 35g
- Selectable resolutions up to 12,000dpi
 - 50 dpi step size from 50 dpi to 10,000dpi
 - 100 dpi step size from 10,100dpi to 12,000dpi
- Enhanced Programmability
 - Low Power Gaming Mode
 - Corded Gaming Mode
- Three-wire serial port interface (SDIO)
- Internal oscillator — no clock input needed
- Customizable response time and downshift time for rest modes
- Angle snapping

Applications

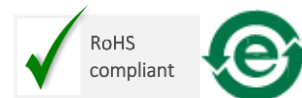
- Wired and wireless Gaming Optical Mouse
- Trackball application

Key Parameters

Parameter	Value
Power Supply Voltage	VDD: 1.80 to 2.10V
Interface	3-wire Serial Peripheral Interface
Supply Current @ VDD = 1.9V Note: includes LED current	Run: 2.3 mA Power Down: 3 μ A
Resolution	Up to 12000 dpi
Tracking Speed	300 ips
Acceleration	35 g
Package Size	9.9 x 12.85 x 6.1 mm ³
Assembled Solution Size (Package Assemble with Lens)	LM31-LNG: 40.7 x 16.0 x 13.9 mm ³ LM33-LSG: 34.5 x 16.0 x 13.9 mm ³ LOAL-LSG1: 33.5 x 16.0 x 13.9 mm ³ LOAJ-LSG1: 15.5 x 12.8 x 10.6 mm ³

Ordering Information

Part Number	Package Type
PAW3311DB-T2MU	8L PDIP
LM31-LNG	Trim Lens
LM33-LSG	Small Trim Lens
LOAL-LSG1	Small Trim Lens
LOAJ-LSG1	Small Form Factor Trim Lens



For any additional inquiries, please contact us at
<http://www.pixart.com>

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1.0 Introduction

1.1 Overview

The PAW3311DB-T2MU is based on Optical Navigation Technology, which measures changes in position by optically acquiring sequential surface images (frames) and mathematically determining the direction and magnitude of movement. The chip contains an Image Acquisition System (IAS), a Digital Signal Processor (DSP), and a three-wire serial port. The IAS acquires microscopic surface images via the lens and illumination system. The images are processed by the DSP to determine the direction and distance of motion. The DSP calculates the ΔX and ΔY relative displacement values. An external microcontroller reads and translates the ΔX and ΔY information from the chip serial port into PS2, USB, or RF signals before sending them to the host PC.

Note: Throughout this document, the PAW3311DB-T2MU is referred to as the “chip”.

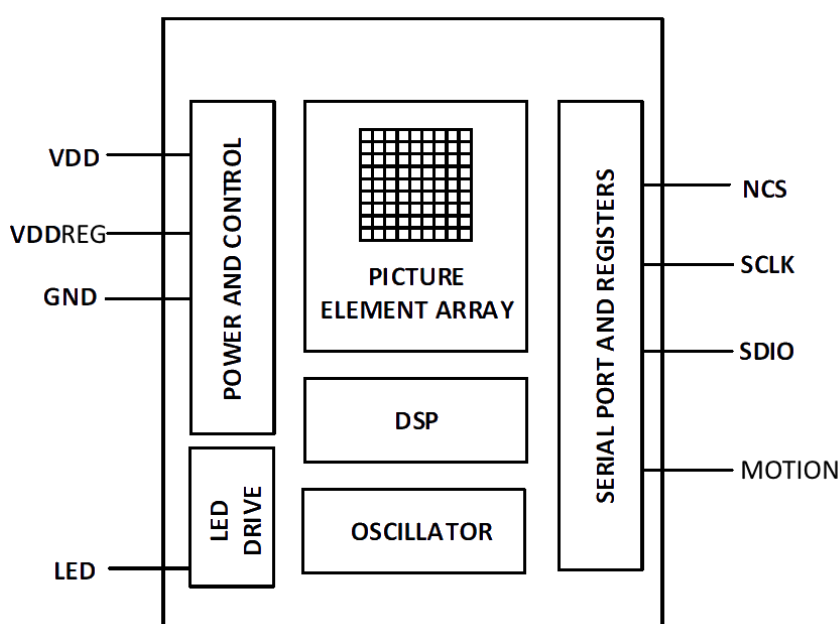


Figure 1. Functional Block Diagram

1.2 Terminology

Term	Description
DSP	Digital Signal Processing
LED	Light Emitting Diode
NCS	Chip Select
VDDREG	LDO output for digital core
VDD	Supply voltage
SCLK	Serial Clock
SDIO	Serial Data In & Out
SPI	Serial Peripheral Interface
GND	Ground
MOTION	Motion Detect

1.3 Pin Description

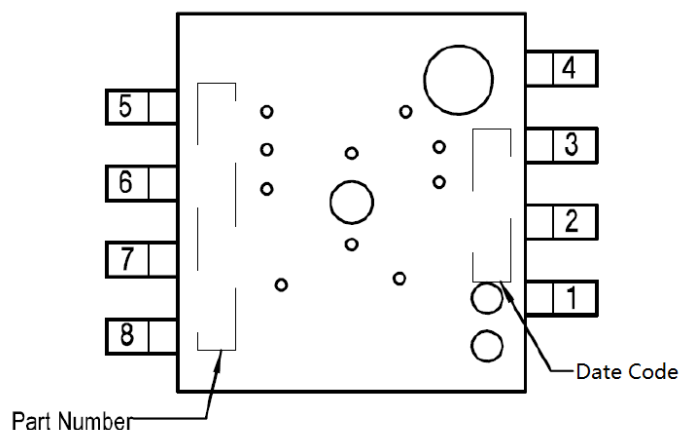


Figure 2. Pin Configuration

Table 1. Pin Description

Pin No.	Name	Type	Description
1	LED	Input	LED illumination control input
2	VDDREG	Power	LDO output (only for sensor internal usage).
3	VDD	Power	Power supply
4	SCLK	Input	Serial clock input
5	MOTION	Output	Motion detection
6	GND	Ground	Ground
7	SDIO	I/O	Serial data in/out
8	NCS	Input	Chip select (active low)

2.0 Operating Specification

2.1 Regulatory Requirement

- Passes FCC “Part15 Subpart, Class B”, “ICES-003:2016 Issue 6, Class B” and “ANSI C63.4:2014” when assembled into a mouse with shielded USB cable using ferrite bead and following PixArt’s recommendations.
- Passes IEC 62471:2006 Photo biological safety of lamps and lamp systems.

2.2 Absolute Maximum Rating

Table 2. Absolute Maximum Rating

Parameter	Symbol	Min.	Max.	Unit	Note
Storage Temperature	T_S	-40	85	°C	
Lead Solder Temperature	T_{SOLDER}		260	°C	For 10 seconds, 1.6mm below seating plane
Supply Voltage	VDD	-0.5	2.1	V	Including V_{NA} (100 mV, peak to peak)
ESD	ESD_{HBM}		2	kV	All pins, Human Body Model
Input Voltage	V_{IN}	-0.5	VDD	V	All I/O pins

Notes:

- Stresses greater than those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. These are the stress ratings only and functional operation of the device at these or any other condition beyond those indicated for extended period of time may affect device reliability.
- The maximum ratings do not reflect eye-safe operation.
- The inherent design of this component causes it to be sensitive to electrostatic discharge. The ESD threshold is listed above. To prevent ESD induced damage, take adequate ESD precautions when handling this product.

2.3 Recommended Operating Condition

Table 3. Recommended Operating Condition

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
Operating Temperature	T_A	0		40	°C	
Power Supply Voltage	VDD	1.8	1.9	2.1	V	Including V_{NA} (100 mV, peak to peak)
Power Supply Rise Time	t_{RT}	0.15		20	ms	0 to Minimum VDD
Supply Noise	V_{NA}			100	mV	10kHz to 75MHz, peak to peak
Serial Port Clock Frequency	f_{SCLK}			8	MHz	Active drive, 50% duty cycle
Supportable USB Polling Frequency	f_{USB}			2	kHz	Corded Gaming Mode
				1	kHz	Low Power Gaming Mode
Distance from Lens Reference Plane to Tracking Surface	Z	2.20	2.40	2.60	mm	
Speed	S				ips	Tested on PixArt standard gaming surfaces at 0, 45 and 90 degree
▪ Low Power Gaming Mode		300				
▪ Corded Gaming Mode		300				
Acceleration	A				g	In run mode PixArt standard gaming surfaces
▪ Low Power Gaming Mode		35				
▪ Corded Gaming Mode		35				
Load Capacitance	C_L			20	pF	SDIO, MOTION
Lift Cut-off Height	Lift		2		mm	LM31-LNG or LM33-LSG
			1			LOAL-LSG1
			1			LOAJ-LSG1

Note: PixArt does not guarantee the chip performance if the operating temperature is beyond the specified limit.

2.4 Thermal Specification

Table 4. Thermal Specification

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
Storage Temperature	T_S	-25	-	80	°C	
Lead-free Solder Temperature	T_P	-	-	260	°C	For 10 seconds, 1.6mm below seating plane for wave soldering

2.5 DC Characteristic

Table 5. DC Electrical Specification

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
DC Supply Current (Run Mode)	I_{DD_RUN}		2.3		mA	- Average Run current - Includes I_{LED} - No load on SDIO & MOTION
- Low Power Gaming Mode - Corded Gaming Mode			3.3			
DC Supply Current (Rest Mode)	$I_{DD_REST1[LPGM]}$ $I_{DD_REST2[LPGM]}$ $I_{DD_REST3[LPGM]}$ $I_{DD_REST1[CGM]}$ $I_{DD_REST2[CGM]}$ $I_{DD_REST3[CGM]}$		800		μA	- Average Rest current - No load on SDIO & MOTION - Includes I_{LED}
Low Power Gaming Mode			30			
			10			
Corded Gaming Mode			800			
			30			
			10			
Power Down Current	I_{PD}		3		μA	
Input Low Voltage	V_{IL}			$0.3 \times V_{DD}$	V	SCLK, SDIO, NCS
Input High Voltage	V_{IH}	$0.7 \times V_{DD}$			V	SCLK, SDIO, NCS
Input Hysteresis	V_{I_HYS}		100		mV	SCLK, SDIO, NCS,
Input Leakage Current	I_{LEAK}		± 1	± 10	μA	$V_{in} = V_{DD}$ or $0V$, SCLK, SDIO, NCS
Output Low Voltage	V_{OL}			0.45	V	$I_{OUT} = 1mA$, SDIO, MOTION
Output High Voltage	V_{OH}	$V_{DD} - 0.45$			V	$I_{OUT} = -1mA$, SDIO, MOTION
Input Capacitance	C_{in}		10		pF	SCLK, SDIO, NCS

Note: All the parameters are tested under recommended operating conditions and measured using HSDL-4261 or LTE-C216-P-W-B-PA illumination sources. Typical values at 25 °C, VDD = 1.9 V and LED current = 24mA.

2.6 AC Characteristic

Table 6. AC Electrical Specification

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
Motion Delay After Reset	$t_{MOT-RST}$	50			ms	From reset to valid motion, assuming motion is present
Shutdown	t_{STDWN}			500	ms	From Shutdown mode active to low current This timing could be affected by Rest3 period
Wake up from Shutdown	t_{WAKEUP}	50			ms	From Shutdown mode inactive to valid motion. Note: A RESET must be asserted after a shutdown. Refer to Section 5.2, also note $t_{MOT-RST}$.
SDIO Output Rise Time	t_{r-SDIO}		6		ns	$C_L = 20pF$

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
SDIO Output Fall Time	$t_{f\text{-SDIO}}$		6		ns	$C_L = 20\text{pF}$
SDIO Output Delay After SCLK	$t_{\text{DLY-SDIO}}$			38	ns	From SCLK falling edge to SDIO output data valid $C_L = 20\text{pF}$
SDIO Output Hold Time	$t_{\text{hold-SDIO}}$	31.25			ns	Data held until next falling SCLK edge
SDIO input Hold Time	$t_{\text{hold-SDIO input}}$	31.25			ns	Amount of time data is valid after SCLK rising edge
SDIO input Setup Time	$t_{\text{setup-SDIO input}}$	31.25			ns	From data valid to SCLK rising edge
SPI Time Between Write Commands	t_{SWW}	5			μs	From rising SCLK for last bit of the first data byte, to rising SCLK for last bit of the second data byte.
SPI Time Between Write and Read Commands	t_{SWR}	5			μs	From rising SCLK for last bit of the first data byte, to rising SCLK for last bit of the second address byte
SPI Time Between Read and Subsequent Commands	t_{SRW} t_{SRR}	2			μs	From rising SCLK for last bit of the first data byte, to falling SCLK for the first bit of data being read.
SPI Read Address-Data Delay	t_{SRAD}	2			μs	From rising SCLK for last bit of the address byte, to falling SCLK for the first bit of data being read.
NCS Inactive After Motion Burst	t_{BEXIT}	500			ns	Minimum NCS inactive time after motion burst before next SPI usage
NCS To SCLK Active	$t_{\text{NCS-SCLK}}$	120			ns	From last NCS falling edge to first SCLK rising edge.
SCLK To NCS Inactive SDIO Read	$t_{\text{SCLK-NCS read}}$	120			ns	From last SCLK rising edge to NCS rising edge, for valid SDIO data transfer
SCLK To NCS Inactive SDIO Write	$t_{\text{SCLK-NCS write}}$	1			μs	From last SCLK rising edge to NCS rising edge, for valid SDIO data transfer
NCS To SDIO High-Z	$t_{\text{NCS-SDIO}}$			500	ns	From NCS rising edge to SDIO high-Z state
Transient Supply Current	I_{DDT}		70		mA	Maximum supply current during the supply ramp from 0V to VDD with minimum 150 μs and maximum 20ms rise time. (Does not include charging currents of bypass capacitors)

Note: All the parameters are tested under recommended operating conditions and measured using HSDL-4261 or LTE-C216-P-W-B-PA illumination sources. Typical values at 25 °C, VDD = 1.9 V & LED current = 24mA.

2.7 External Illumination Source Requirement

LM31-LNG, LM33-LSG and L0AL-LSG1 are designed to be used with an external 5mm IR illumination source and L0AJ-LSG1 is designed to be used with an external SMT IR illumination source. The lens provides the necessary guided illumination for optimum chip operation.

2.7.1 5mm IR Illumination Source Requirement

Parameter	Symbol	Min.	Nominal	Max.	Unit	Note
Viewing Angle	θ	-	26	-	degree	$I_{FDC} = 20mA$
Radiant Intensity	I_E	-	36	-	mW/sr	$I_{FDC} = 20mA$
		-	180	-	mW/sr	$I_{FDC} = 100mA$
Package Type	-	-	-	-	-	Standard 5mm LED

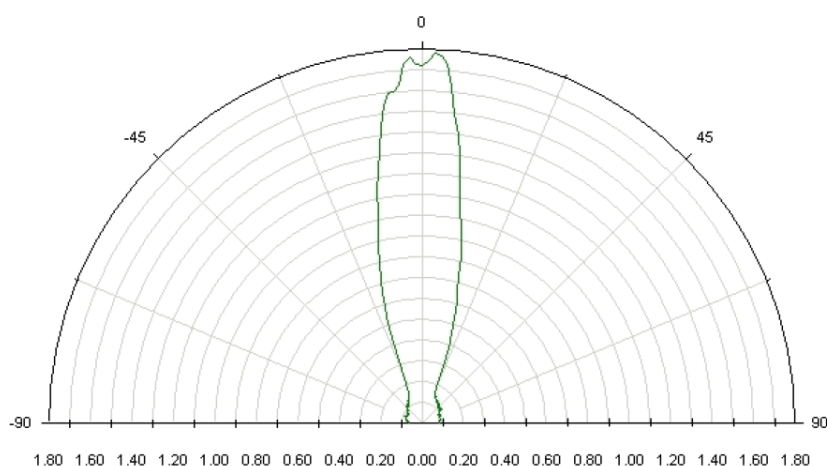


Figure 3. 5mm IR Illumination Source Radiation Characteristic

Note: Do contact PixArt for the illumination source recommendation.

2.7.2 SMT IR Illumination Source Requirement

Table 7. SMT IR Illumination Source Requirement

Parameter	Symbol	Min.	Nominal	Max.	Unit	Note
Viewing Angle	θ	-	100	-	degree	
Radiant Intensity	I_E	2.2	3.0	-	mW/sr	$I_{FDC} = 20\text{mA}$
Package Type	-	-	-	-	-	1206 package

Note: Position of LED die within the illumination source package must be controlled within $\pm 0.1\text{mm}$ to minimize variation in the output radiation pattern.

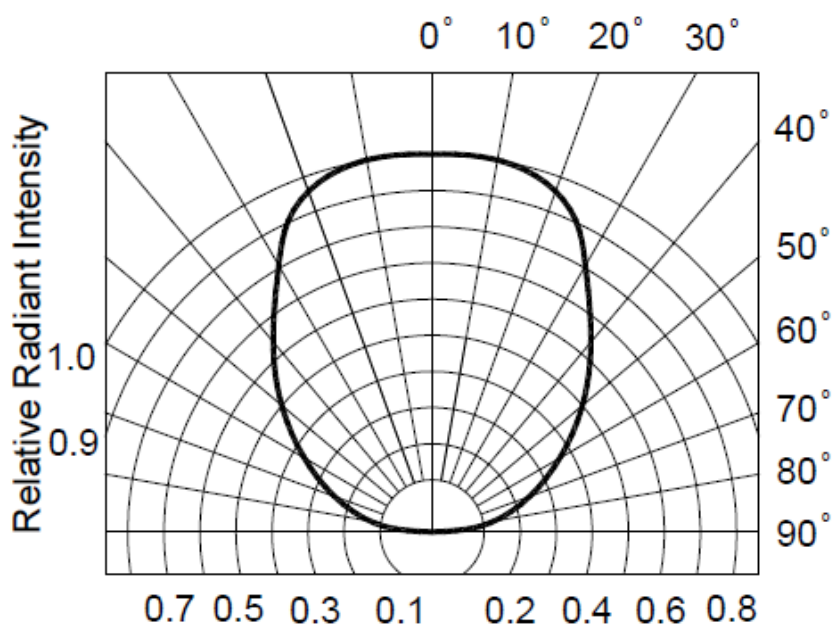


Figure 4. SMT IR Illumination Source Radiation Characteristic

Note: Do contact PixArt for the illumination source recommendation.

3.0 Mechanical Specification

3.1 Mechanical Dimension

Table 8. Package Dimension

Parameters	Nominal	Min.	Max.	Unit
Package Body Dimension X	9.90	9.80	10.00	mm
Package Body Dimension Y	9.10	9.00	9.20	mm
Package Width (inclusive pins)	12.85	12.35	13.35	mm
Lead Length	5.15	5.05	5.25	mm
Lead Pitch	2.00	1.85	2.15	mm
Total Lead Count	8	-	-	-
Lead Offset	1.00	-	-	mm
Lead Width	0.50	0.40	0.60	mm
Hole Diameter	0.70	0.65	0.75	mm
Center of Hole from edge of body X	4.55	4.45	4.65	mm
Center of Hole from edge of body Y	3.92	3.82	4.02	mm

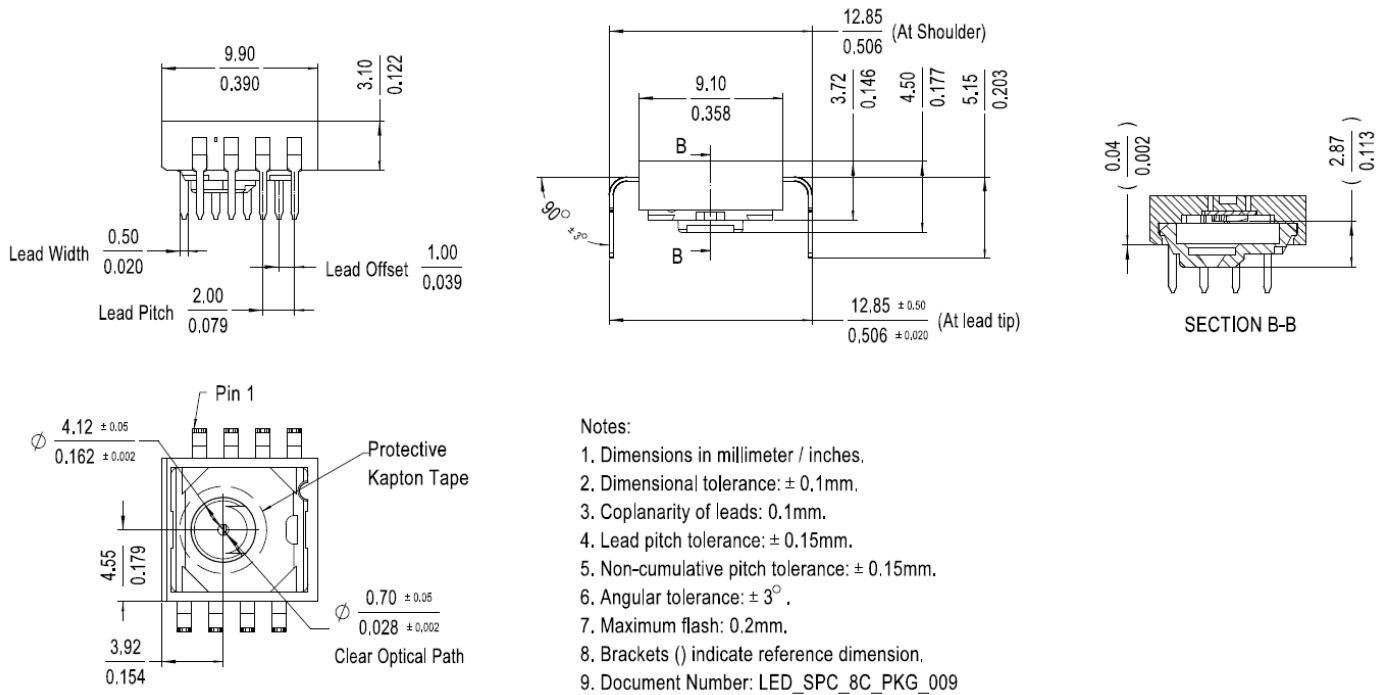


Figure 5. Package Drawing Outline

3.2 Assembly Drawing

3.2.1 Recommended PCB Footprint

It is highly recommended to follow the chip orientation in Figure 4 and Figure 5 to achieve optimum tracking performance.

Table 9. Recommended PCB Footprint

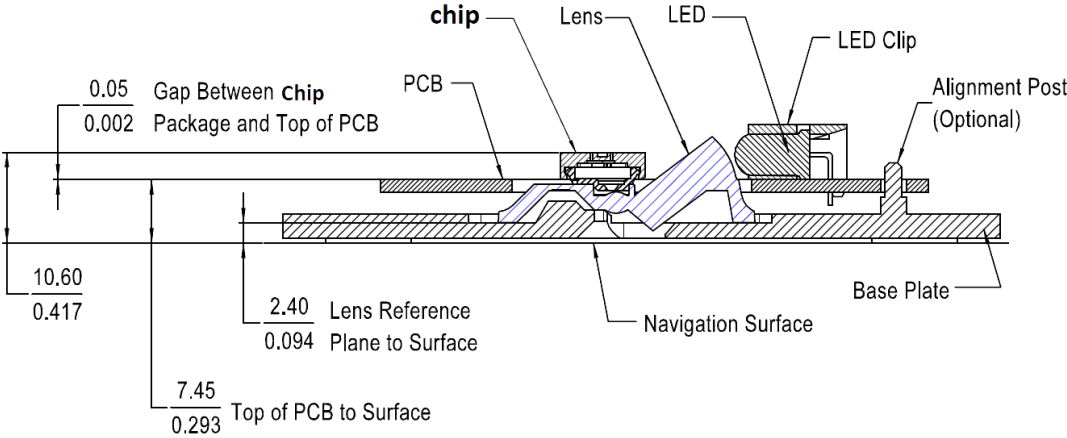
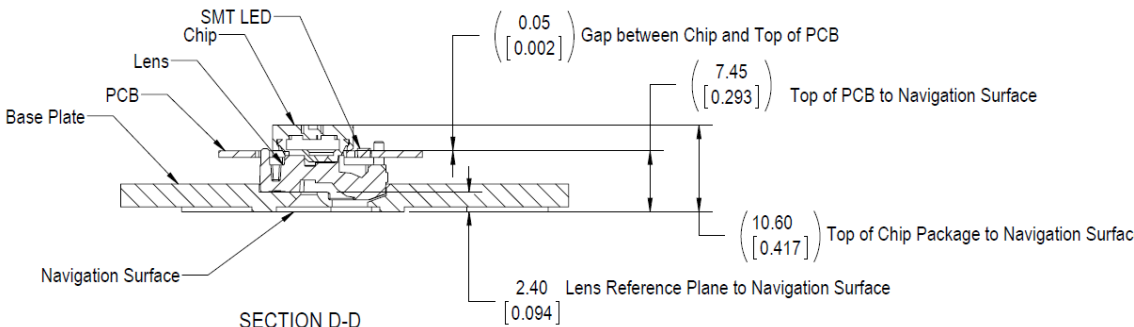
Drawing		Lens
Front of Mouse Buttons: 2 Pin #1 Optical Center Clear Zone 2X $\phi 0.80$ [0.031] 3X $\phi 3.00$ [0.118] Optional hole for alignment post if used Notes:		LM31-LNG, LM33-LSG and LOAL-LSG1
Front of Mouse Buttons: 2 14X R0.40 Max LED Soldering Pad R0.90 X 2.00 LED Cut Out LED Center Clear Zone 8X $\phi 0.80$ PCB Thickness 0.80mm		LOAJ-LSG1

Note:

1. Dimension in millimeters
2. View from component side of PCB (or Top view of the mouse)

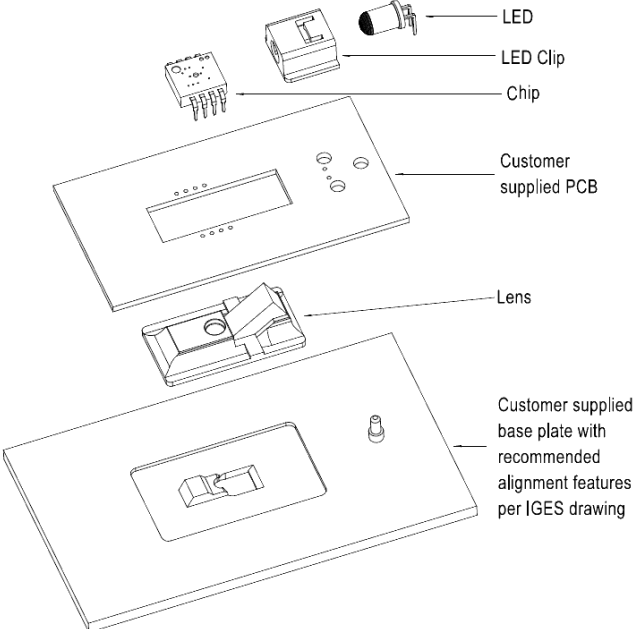
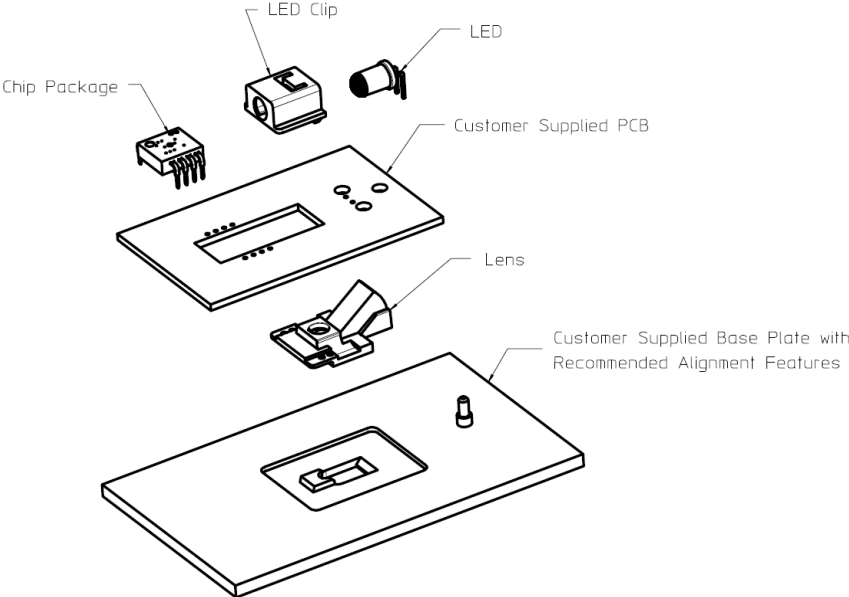
3.2.2 Cross-sectional View

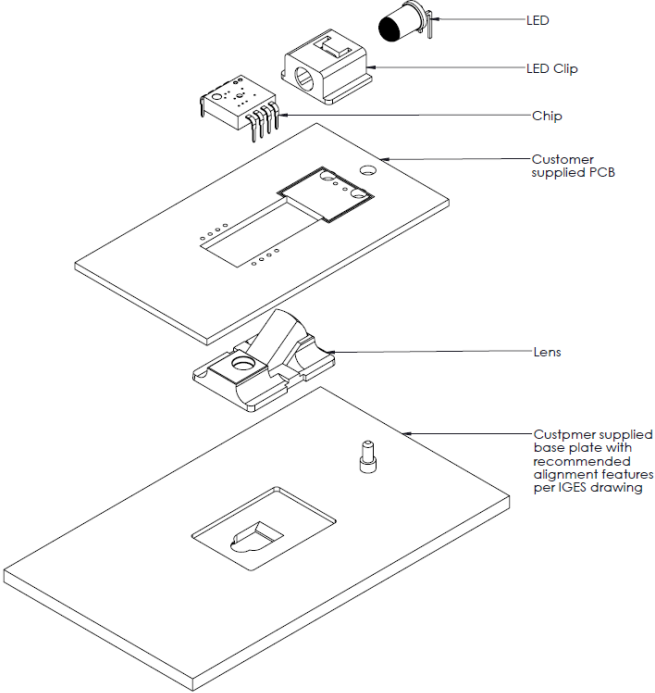
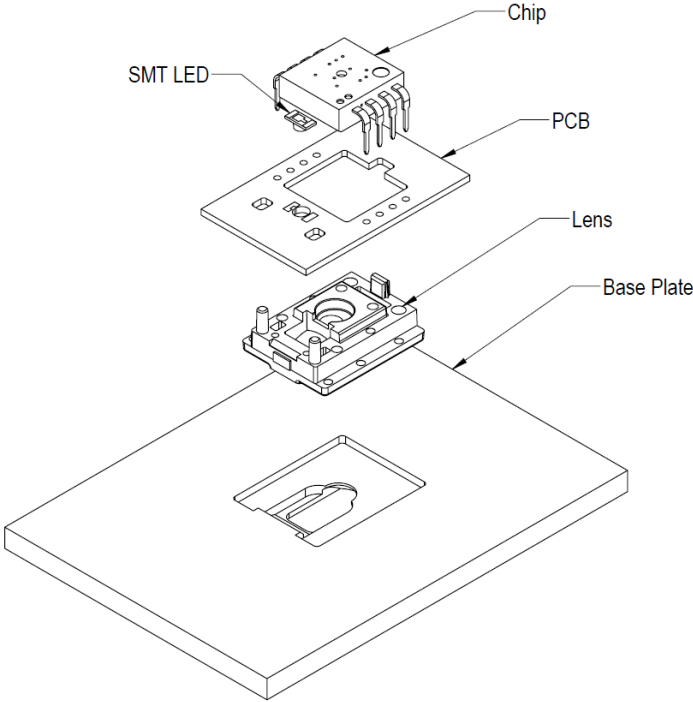
Table 10. Cross-sectional View

Drawing	Lens
 Note: 1. Dimension in millimeter/inches 2. Pin 1 of the chip needs to be located near to the LED 3. The LED must be bent 90 degree with LED flange touching the PCB	LM31-LNG, LM33-LSG and LOAL-LSG1
 Note: 1. Dimension in millimeter [inch]. 2. General dimensional tolerance: $\pm 0.10\text{mm}$. 3. Bracket () indicates reference dimension.	LOAJ-LSG1

3.2.3 Exploded View

Table 11. Exploded View of Assembly

Drawing	Lens
 LED LED Clip Chip Customer supplied PCB Lens Customer supplied base plate with recommended alignment features per IGES drawing	LM31-LNG
 LED Clip LED Chip Package Customer Supplied PCB Lens Customer Supplied Base Plate with Recommended Alignment Features	LM33-LSG

Drawing	Lens
	LOAL-LSG1
	LOAJ-LSG1

Note: Refer to the Lens datasheet for the detail.

3.3 Package Marking

Refer to Figure 2. Pin Configuration

Table 12. Code Identification

Code	Marking	Description
Product Number	PAW3311DB-T2MU	Part number label
Lot Code	YYWWXXXXX	YYWW=Date code XXXXX= PixArt Reserved

4.0 System Level Description

4.1 Reference Application Schematic Diagram

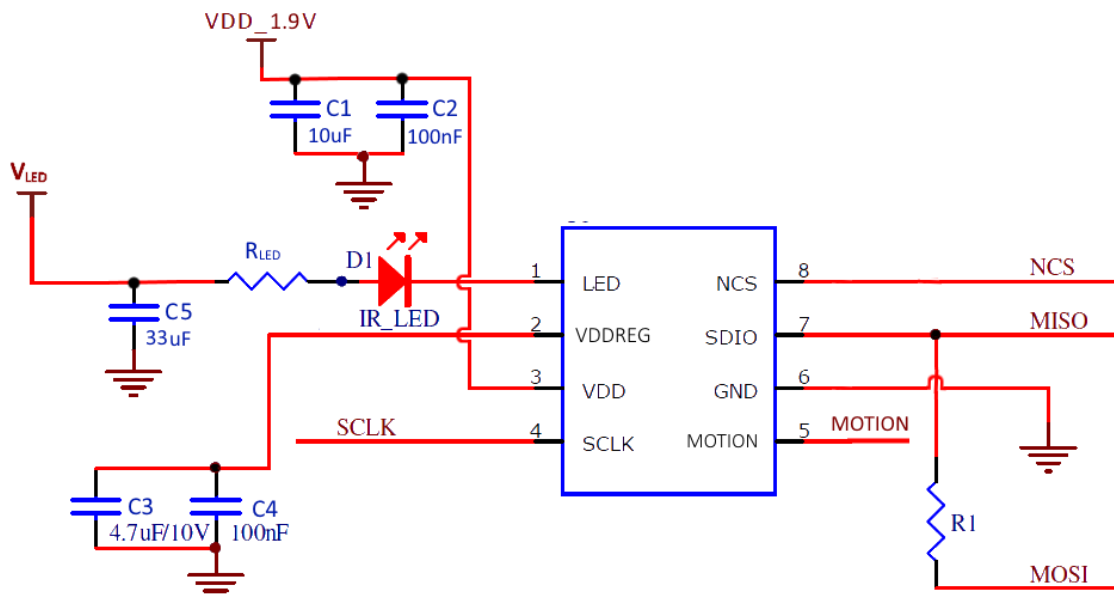


Figure 6. Reference Mouse Schematic

Table 13 shows the recommended value of R_{LED} and V_{LED} to obtain 24mA current for the PixArt recommended illumination source. Recommend to use R_{LED} with 1% tolerance.

Table 13. Recommended R_{LED}

V_{LED} (V)	Recommended R_{LED} (Ω)
1.9	18

Table 14 shows the recommended value of R1 for 2MHz, 4MHz and 8MHz of serial port clock frequency. Recommend to use R1 with 1% tolerance.

Table 14. Recommended R1

Serial Port Clock Frequency (MHz)	Recommended R1 (Ω)
2	3.3k
4	1.0k
8	240

4.2 PCB Assembly Recommendation

1. Assemble and reflow the SMT LED and other components onto the PCB. (Only applicable for LOAJ-LSGI lens with SMT illuminations source, skip this step for other lens)
2. Insert the integrated chip and all other electrical components into PCB.
3. Wave-solder the entire assembly in a no-wash solder process utilizing solder-fixture. A solder-fixture is required to protect the chip from flux spray and wave solder.
4. Avoid any solder flux onto the chip body as there is potential for flux to seep into the chip package, the solder fixture must be designed to expose only the chip leads to flux spray & molten solder while shielding the chip body and optical apertures. The fixture needs to set the chip at the correct position and height on the PCB.
5. Place the lens onto the base plate. Avoid contamination on the optical surfaces.
6. Remove the protective Kapton tapes from optical apertures of the chip. Avoid contaminants from entering the apertures. Do not place the PCB with the chip facing up during the entire mouse assembly process. Hold the PCB vertically when removing Kapton tape.
7. Insert PCB assembly over the lens onto the base plate, aligning the guide posts on the lens to the PCB. The chip package will self-align to the chip package via the guide posts. The optical position reference for the PCB is set by the base plate and lens. Note that the PCB motion (due to button presses) must be minimized to maintain optical alignment.
8. Recommendation for LOAJ-LSG1 lens: The lens can be permanently secured to the PCBA by melting the lens' guide posts over the PCB with heat staking process.
9. Install mouse top case. There must be a feature in the top case to press down onto the PCB assembly to ensure all components are stacked or interlocked to the correct vertical height.

Revision History

Revision Number	Date	Description
1.0	15 Nov 2022	Initial Release